



Automation and Robotics Engineering

Nonlinear Dynamics and Control: Final project

Analysis and Control of Love Dynamics in "Gone with the Wind"

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1. Control problem formulation

Closed-loop system:

$$\begin{cases} \dot{x}_1(t) = -x_1(t) + kx_2(t)e^{-x_2(t)}, & x_1(0) = 1, \\ \dot{x}_2(t) = -x_2(t) + kx_1(t)e^{-x_1(t)} + u(t), & x_2(0) = 1.5, \\ y(t) = h(x(t)) = x_1(t) & k=15 \end{cases}$$

where $x_1(t)$ and $x_2(t)$ represent, respectively, Scarlett's and Rhett's feelings.

Control objectives:

- Asymptotic tracking (main objective):

$$e_{ss} = \lim_{t \rightarrow +\infty} (y_d - y(t)) = \lim_{t \rightarrow +\infty} (P_{2x} - x_1(t)) = 0$$

- Maximum overshoot percentage: $S\% < 5\%$
- Maximum control effort: $u_{\max} = \max_{t \in [0, +\infty]} |u(t)| < 5$

2. Open-loop analysis

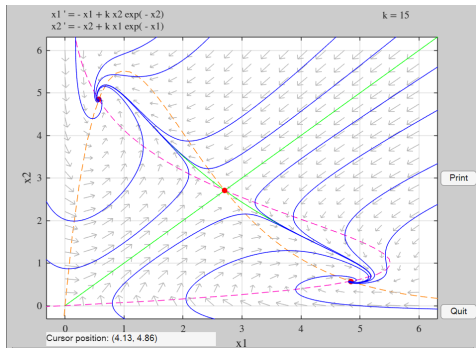
Equilibrium points:

- $P_1 \approx (0.57, 4.84)$: stable spiral
- $P_2 \approx (2.7, 2.7)$: saddle point
- $P_3 \approx (4.84, 0.57)$: stable spiral

Limit cycles:

- No limit cycles appear in the domain of interest (Dulac's criterion):

$$\nabla \cdot f = -2 < 0$$



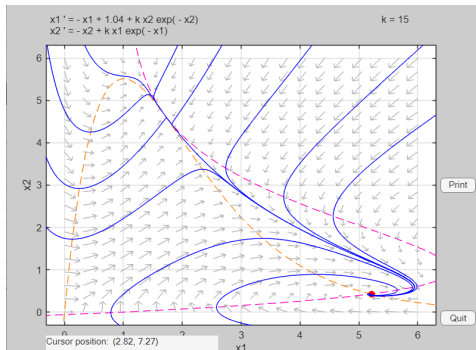
3. Bifurcation analysis

Bifurcation parameter:
 A_2

- A_2 increases $\Rightarrow P_1$ and P_2 move closer.
- $A_2 \approx 1.037 \Rightarrow P_1$ and P_2 collide and disappear.
- **Saddle-node bifurcation** occurs.

After the bifurcation:

- Only one globally stable equilibrium remains.

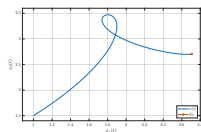


5. Synthesis

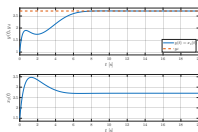
5.1. Linear controller

Controller parameters:

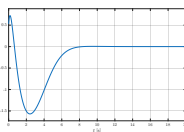
$$K = [2.4809 \quad -1.3], \quad k_i = -0.7728$$



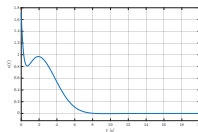
Phase plane



States



Control input



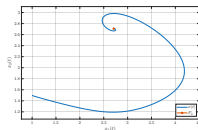
Output error

- Asymptotic tracking achieved: $e(t) \rightarrow 0$ ✓
- Overshoot: $S\% = 0.12\% < 5\%$ ✓
- Control effort specification respected: $u_{\max} < 5$ ✓
- The whole state converges to the desired saddle point P_2 ✓

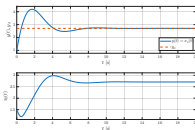
4.2. Input-Output feedback linearization controller

Controller parameters:

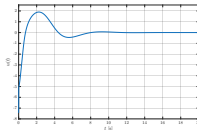
$$k_1 = k_2 = 1$$



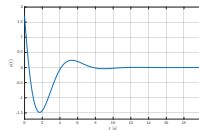
Phase plane



States



Control input



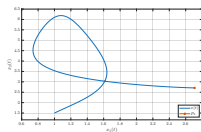
Output error

- Asymptotic tracking achieved: $e(t) \rightarrow 0$ ✓
- Overshoot: $S\% = 54.68\% > 5\%$ ✗
- Maximum control effort not respected: $u_{\max} = 7.44 > 5$ ✗
- The whole state converges to the desired saddle point P_2 ✓

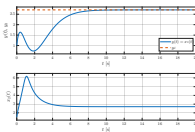
4.3. Input-State feedback linearization controller

Controller parameters:

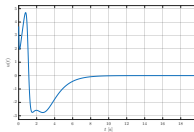
$$K = \begin{bmatrix} -3.62 & -3.3 \end{bmatrix}, \quad k_i = 1.32$$



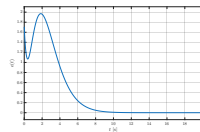
Phase plane



States



Control input



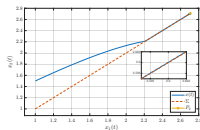
Output error

- Asymptotic tracking achieved: $e(t) \rightarrow 0$ ✓
- No overshoot ✓
- Control effort specification respected: $u_{\max} = 4.7066 < 5$ ✓
- The whole state converges to the desired saddle point P_2 ✓

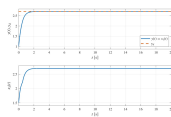
4.4. Sliding mode controller

Controller parameters:

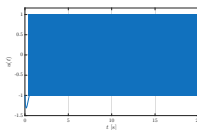
$$k_{\text{law}} = 1, \quad p_1 = 1, \quad p_2 = -1$$



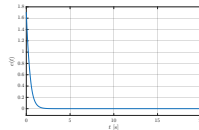
Phase plane



States



Control input



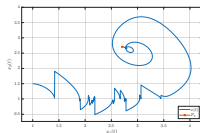
Output error

- Asymptotic tracking achieved: $e(t) \rightarrow 0$ ✓
- No overshoot ✓
- Lowest control effort: $u_{\text{max}} = 1.315 < 5$ ✓
- Chattering phenomenon due to the discontinuous control law ✗

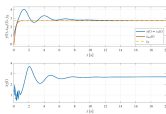
4.5. MCS-based control of an IS feedback linearized system

Controller parameters:

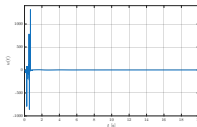
$$Q = I_2, \quad \beta = 1, \quad \alpha = 10, \quad A_m = \begin{bmatrix} 0 & 1 \\ -64 & -16 \end{bmatrix}, \quad B_m = \begin{bmatrix} 0 \\ 64 \end{bmatrix}$$



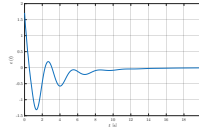
Phase plane



States



Control input



Output error

- Tracking problem solved: $e(t) \rightarrow 0$ ✓
- Overshoot: $S\% \approx 48\% > 5\%$ ✗
- Very large control effort: $u_{\max} = 1311 \gg 5$ ✗
- The whole state converges to the desired saddle point P_2 ✓

5. Robustness analysis

**Perturbed
plant:**

$$\begin{cases} \dot{x}_1 = -x_1 + kx_2 e^{-x_2} + d_1(t), & x_1(0) = 1, \\ \dot{x}_2 = -x_2 + kx_1 e^{-x_1} + d_2(t) + u, & x_2(0) = 1.5, \\ y = x_1. \end{cases}$$

- **SFC**: rejects constant step disturbances on $y(t)$ thanks to integral action, but not white noise.
- **IOFBL**: poor robustness, due to exact cancellation and strong model dependence.
- **ISFBL**: similar to SFC; integral action ensures tracking only for step disturbances.
- **SMC**: robust with respect to matched disturbances entering through the control channel.
- **MCS + ISFBL**: applicable to the nonlinear plant, but inherits feedback-linearization model dependence.

7. Conclusions

The open-loop love dynamics naturally converge toward unbalanced emotional configurations, but appropriate nonlinear control strategies can drive the system toward a balanced relationship, represented by P_2 .

- **SFC**: good nominal performance and all specifications satisfied.
- **IOFBL**: tracking achieved, but overshoot and control effort are too large.
- **ISFBL**: all specifications satisfied.
- **SMC**: best nominal and robustness performance, but affected by chattering.
- **MCS + ISFBL**: tracking achieved, but high overshoot and control effort.

The **Sliding Mode Controller** is the most effective solution, although practical implementation would require chattering mitigation.